# Mother-Infant Separation in Rhesus Monkeys as a Model of Human Depression

## A Reconsideration

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• **Nineteen rhesus monkeys between the ages of 5.9 and 8.5 months were separated from their mothers in five different studies. While in two of the studies, data indicated behavioral responses roughly parallel to Bowlby's protest-despair response to maternal separations, data across all five studies were sufficiently variable to bring this technique into serious question as a reliable and predictable animal model for neurobiologic and rehabilitative studies.**

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Separation-induced depression in human children has been studied by several investigators since the syndrome of anaclitic depression was first described by Spitz[1] in 1947. For example, Robertson and Bowlby[2] postulated that the reaction of infants and children to maternal separation occurred in distinct stages. The initial "protest" stage was characterized by high agitation and attempts to return to mothers, the subsequent "despair" stage by weeping and reduced activity, and the final "detachment" phase by withdrawal from any activity in the environment, and indeed by rejection of the mother's initial attempts to reestablish the relationship in those cases when the mother returned.

The studies of Jensen and Tolman[3] and Seay and associates,[4] which reported strikingly similar responses to maternal separation in infant rhesus monkeys, suggested the possibility that a behavioral parallel existed in monkeys that might provide a model of human depression. Such a model would allow investigation of the neurobiologic substrates of affective disorders in a manner ethically precluded in human subjects.

The value of such a model, however, clearly rests on the predictability and stability of the behavioral state induced in the monkey. We shall review the literature on maternal separation in rhesus monkeys and present data from our laboratory that indicate that this model is not sufficiently predictable or stable for use in neurobiologic or rehabilitative studies.

Laboratory research concerning mother-infant separation in nonhuman primates began with the study of Jensen and Tolman,[3] in which the authors reported results of the separation of two infant pig-tailed macaques (*Macaca nemestrina*) from their mother for brief periods at 7 and 5 months of age, respectively. Their separation procedure consisted of separating the infants from their mothers, placing them in an observation cage, either with their own or the strange mother, and then reseparating and reuniting them with the other mother in the same cage. Thus, their separation was for brief periods, and the study was directed toward specificity of mother-infant relationships. The authors reported that in this very brief separation-reunion paradigm, the infants and mothers showed high levels of arousal and distress. When the infants were reunited with their own mothers, they immediately showed high levels of clinging and similar mother-directed behaviors. However, when reunited with strange mothers, although the infants attempted to establish physical closeness, the strange mother rejected their attempts. The authors conclude that: (1) "own infant"-directed behavior on the part of the mothers was higher than "other infant"-directed behaviors after brief separation; (2) mother-directed behaviors are increased in the infant after brief separation; (3) learned own-mother specificity occurs after several separations; and (4) that separation increases the interactive behavior between mother and infant.

Seay and associates[4] studied the effects of separation on four mother-infant pairs. The infants were separated from their mothers in pairs, by means of a Plexiglas panel, into a common playpen. Thus, while mother-infant interaction was denied, peer interaction was available during the separation. The authors reported that the infants showed high levels of distress behaviors during the first week of the separation period, including increased locomotion, distress vocalizations, and attempts to return to the mother. Throughout the three-week separation period,

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the infants showed dramatically decreased levels of active-play behaviors as compared to the preseparation period. The authors interpret their data as supporting the Bowlby formulation of protest, despair, and withdrawal, with the exception that the behavior consistent with Bowlby's description of withdrawal was seen in only one of the four infants.

In a similar study, Seay and Harlow[5] separated two groups of four infants from their mothers into a pen. The infants had access to each other during daily one-half-hour testing sessions, while the mothers were removed to another room for the duration of the two-week separation period. The authors reported an initial protest stage consisting of high levels of locomotion and distress vocalizations. During the period of separation, the level of self-mouthing and social-play behaviors decreased, and although the authors do not report the specific time course for the change from increased vocalization-locomotion to despair behaviors, they do note such a change. In addition, the authors note that in contrast to the earlier study when visual contact with the mother was available, in this study, where the mother was completely removed, the protest exhibited by the infants seemed less intense, although no supporting data are presented for this conclusion.

Hinde and co-workers[6] removed, for six days, the mothers of four infants (8 months of age) living in a pen. They found that the infants exhibited a similar response to the separation as that shown by the infants in the cited studies[3-5]—an initial increase in distress vocalizations and locomotion that persisted for one or two days. Subsequently, the infants showed lowered levels of locomotion and vocalization, and the authors reported a decrease in social and manipulative play through the course of the six-day separation. On return of the mothers, the infants showed increased levels of mother-directed behavior as compared with the preseparation period.

Kaufman and Rosenblum[7-9] have studied the effects of mother-infant separation in four pig-tailed macaques. When the infants were aged 5 to 6 months, the mothers were removed from the pen, where the animals had been reared from birth. The separations of the infants took place at different times, with some overlap of separations between subjects. The authors reported a clear "agitation" stage, lasting 24 to 36 hours, in all subjects. This was followed by a "depression" phase, lasting an additional five to six days in three of the four subjects, and a "recovery" phase lasting through the remainder of the month-long separation in all four subjects. Agitation was marked by distress vocalizations, high levels of locomotion, and increased digit sucking. Depression consisted of decreases in activity, increase in huddled nonlocomotion behavior, and what was subjectively described as the withdrawn characteristics of their behavior. Recovery consisted of lessening of the depression behaviors and increases in more normal patterns of peer play and locomotion. Although the subjective and objective behavioral depression continued, it appeared to be abated. The authors interpret their data as being consistent with a possible primate model of Spitz's "anaclitic depression," and postulate a system of conservation-withdrawal consistent with reports by Engel.[10,11]

In contrast to data obtained in the previous study with pig-tailed and bonnet macaques. Specifically, bonnet monthat bonnet macaques (*Macaca radiata*) subjected to similar separations showed less severe reactions to separation than their pig-tailed subjects. They postulate this difference to be due to basic differences in the social patterns of pig-tailed and bonnet macaques. Specifically, bonnet monkeys seem to have much higher levels of "passive-contact" behaviors than do pig-tailed monkeys, and as a result, young bonnet macaques have greater opportunity during early life to form relationships with monkeys other than their own mothers. In fact, during the separations, the bonnet macaque infants established substitute mother-like relationships with other adults when their own mothers were removed. This replacement of the mother perhaps explains why the authors found no behavior corresponding to the depression they found in the pig-tailed infants.

Spencer-Booth and Hinde[12] reported the effects of removal for six days of the mothers of four 8-month-old infants from stable pen-housed groups of one adult male and four or five adult female rhesus monkeys. They noted that the separation was accompanied by increases in vocalizations and a transient increase in locomotion, although the locomotion was described as less vigorous than preseparation locomotion. After the first day, locomotion scores dropped precipitously to below preseparation levels. The authors describe the change in locomotion, along with decreased levels of social interaction and environmental manipulation, as being a depression, and suggest that their data, while showing no "detachment," agree with Bowlby's[13] formulation of protest, despair, and detachment responses to maternal separation. In addition, the authors discussed the behaviors of the mother-infant pairs after reunion, and note that mother-infant contact increased over preseparation levels, the largest increases being seen in those mother-infant pairs who had highest contact scores before separation.

Spencer-Booth and Hinde[14] again studied the effects of mother removal for six days in 19 rhesus monkey infants, this time when the infants were at ages 18, 21, 25, and 30 weeks. They noted that certain of their subjects were separated at two of the ages listed. After comparing data from the various groups, they found that age at separation made no significant difference in the data. They also found that repetition of separation made no difference at a given age. They did find, as in their previous study, that infants at all ages showed depression during the six days of separation. They reported no clear increase in locomotion immediately following separation, and reported some cases where locomotion sharply decreased "straight away." In addition to depression of locomotor activity, the usual suppression of play and environmental manipulation was also noted. On reunion of mothers and infants, there were increased levels of contact between mother and infant, when compared to preseparation levels. The authors note two caveats, however. First, they found a high degree of variability in behavior of the infants, and second, they caution against projecting the meaning of their data onto studies of longer-term separations.

Suomi et al[15] compared the responses of infant rhesus



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Outline of Experimental Protocols

| Experimental Protocols | Experiment 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Separation subjects | 5 males<br>1 female | 2 males<br>1 female | 3 males | 4 males | 4 females |
| Age at separation, mo (mean ± SD) | 5.9<br>±0.2 | 6.3<br>±0.1 | 7.9<br>±0.92 | 7.1<br>±1.0 | 8.5<br>±0.5 |
| Total length of separation | 6 days | 6 days | 10 days | 5 wk | 3 wk |
| Type of data taken | Clock-and-counter, modified frequency | Clock-and-counter, modified frequency | Modified frequency | Clock-and-counter | Clock-and-counter |
| Rearing prior to separation | Single with mother | Single with mother | Single with mother | Pen with other separation subjects and mothers | Single with mother, 3 subjects; single with mother and father, 1 subject |
| Type of separation | Separate room, see and hear other subjects | Separate rooms, see and hear other subjects | Adjacent cage, separated from mother by Plexiglas | 2 weeks by Plexiglas, 2 weeks by steel, 1 week with mother gone | 1 week by Plexiglas, 18 hr together, 1 week by steel, 18 hr together, 1 week with mother gone |

monkeys to permanent separation from their mothers at ages 60, 90, and 120 days. On separation, half of the four animals in each group were singly caged, while the other two were housed together in cages twice as large. The authors found no differences in preseparation behavior in any of the groups. However, responses to separation varied between the various age groups, and between pair-and singly-caged conditions following separation. In general, 90-day-old infants showed the severest reaction to separation, and singly-housed subjects reacted more intensely than pair-housed subjects. The authors reported more active disturbance in days 1 and 2 than in days 3 through 7 after separation, more passive disturbance in days 1 and 2 than in days 3 through 7, and interactions of both of the above with housing condition and age. Significantly ($P < .05$), more self-clasp huddling occurred during days 3 through 7 than days 1 and 2, and this main effect also interacted with differences of age and housing condition. Thus, their data roughly fit the protest-despair model of response to mother-infant separation, but in contrast to Spencer-Booth and Hinde,[14] they report differences in response to separation at different ages. The methodologies of the two studies are sufficiently different to contribute to the different findings, in addition to the fact that the subjects of Spencer-Booth and Hinde were, on the whole, older—4 to 8 months of age as compared with 2 to 6 months in this study.

In the studies described above, with the exception of the study of bonnet macaques, the response to mother-infant separation roughly fits the model of protest and despair, usually in time periods within six days of separation. We will present new data from five studies of mother-infant separation in the rhesus monkey at ages 5.9 to 8.5 months of age that we believe do not support a simple protest-despair response model of mother-infant separation.

A total of 20 rhesus monkeys were used as separated subjects in the five studies to be presented, and an additional ten animals were used as biochemical and behavioral control subjects for the various studies (Table). Excluding one study by Spencer-Booth and Hinde,[14] this is more subjects than were previously studied in the entire primate separation literature.

## MATERIALS AND METHODS

### Experiment 1

One female and five male rhesus monkeys ranging in age from 4½ to 5 months of age were used as subjects, and two males and two females of the same age were used as control animals. Prior to separation, all subjects were housed individually with their mothers in one room. Behavioral observations were done twice daily for five minutes on each infant at noon and 4 PM, using simultaneous clock-and-counter and modified frequency scoring systems by two observers. After four weeks of this testing protocol, the six experimental animals were separated from their mothers who were returned to the breeding colony. The infants were taken to another room where they were placed in individual cages (60×60×45 cm) adjacent to one another. There were no other animals in the separation room. Thus, while separated, the subjects had visual and auditory contact with one another, but no other contact was available. Control subjects remained undisturbed in their cages for the duration of the study.

Observations on the six separated subjects were taken four times daily, at 7 AM, noon, 4 PM, and 9 PM, again with two observers simultaneously recording behaviors with modified frequency and clock-and-counter scoring systems. After six days of observation in this manner, all ten subjects were killed for biochemical data by means of a spinal cord section under pentobarbitol sodium anesthesia. More detailed methodology and results from this study have been presented elsewhere.[16]

### Experiment 2

Separation subjects were one female and two male rhesus monkeys between 5 and 5½ months of age at time of separation. Control subjects were two males and one female of similar ages. Protocol for this experiment was identical in all respects to experiment 1, except that only six days of preseparation observations were performed, instead of 20 days.

### Experiment 3

Separation subjects were three male rhesus monkeys 8 to 9 months of age. These subjects had been reared with their mothers from birth in single cages and were observed, using the same modified frequency scoring system described above, in cages similar to their rearing cages (70×70×90 cm) for two weeks prior to separation, four times each day at 7 AM, noon, 4 PM, and 9 PM. The



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subjects were separated from their mothers after two weeks and placed in the cage next to the one in which they had been living, separated from the adjacent cage containing their mother by 0.6 cm of clear Plexiglas. Observations were continued on the same schedule, and subjects' activity was automatically monitored by parallel photocells dividing the cage into thirds, 10 cm from the floor. After 9½ days of observation, the subjects were killed in a similar manner to the two preceding groups. Three biochemical control animals of approximately the same age were killed at the same time, in randomized order.

### Experiment 4

Separation subjects were four male rhesus monkeys, aged 6½ to 8½ months at time of separation. They had been reared as a group with their mothers, in a pen (2×2×3 meters), since birth. Thus, this particular group, in contrast to all others reported here, had peer social contact available all their lives. Control subjects were four female rhesus monkeys, of the same age as those in the experimental group. All subjects were observed four times daily, at 7 AM, 11 AM, 4 PM, and 9 PM, 5 days per week prior to separation, and 7 days per week following separation, using a clock-and-counter scoring system of 11 behavioral categories. Separation subjects were removed from their pens and separated from their mothers by standard laboratory procedures. Mother-infant pairs were then placed in individual cages (70×70×9 cm) adjacent to each other, separated by 0.6 cm of Plexiglas. Thus, as in experiment 3, the infants could see only their own mothers, but had auditory contact with all other mother-infant separated pairs. The control subjects were left undisturbed. Photocells mounted on the infants' cages, as in experiment 3, monitored the subjects' activity 24 hours each day. After 14 days in this situation, the Plexiglas was replaced by a steel separation panel, so that visual contact between mother and infant was no longer possible. (One subject managed to achieve access to his mother's cage for approximately 16 hours at this time by moving the steel panel aside enough to squeeze through into his mother's cage. The two were reseparated and the cage defect repaired as soon as possible.) Following 14 days of living in this separated condition, all the mothers were returned to the breeding colony, and observations of the infants were continued for seven days, 4 times per day. At this time, the study was terminated.

### Experiment 5

Subjects were four female rhesus monkeys, the control group of experiment 4, whose mean age was 8.5 months at the time of separation from their mothers. Protocol for this experiment was identical to experiment 4, with two exceptions. First, each stage of the separation lasted only seven days (in contrast to 14 days for the first two stages of experiment 4), and second, all subjects were allowed to return to their mothers overnight (ten hours), between each of the stages (at the end of seven and 14 days). Subjects were reseparated at 6:30 AM the following morning, one-half hour before the first observation period of the subsequent phase of the study. Observation schedules, type of separation, photocell activity, monitoring, and scoring systems were identical for experiments 4 and 5.

### Observational Scoring

Observation was performed in all experiments on the infant only. Subjective records were kept of the behaviors of the mothers, but will not be included here. Two basic systems were used for data acquisition: (1) clock-and-counter (absolute frequency and duration) and (2) modified frequency. All observations were five minutes long for each animal at each observation time, in all studies.

Clock-and-counter scoring systems utilized an apparatus consisting of a manually operated keyboard connected to a set of total-duration clocks and absolute-frequency counters, such that when a key representing a given behavior was pressed, the counter registered one count, and simultaneously, the appropriate clock started accumulating time in seconds. Thus, the number of occurrences of all behaviors during the five-minute period, as well as the total time spent in those behaviors, was available and constituted the data for that five-minute session. In all cases, the scoring system was exhaustive (at least one behavior being scored at all times), but not mutually exclusive. The behavior categories scored using this system consisted of ten behaviors scored for frequency and duration, and in experiments 4 and 5, absolute frequency only of all vocalizations. Behavioral categories scored in experiments 1 and 2 are as follows: ventral contact; nonventral contact; locomotion; environment exploration; huddling; rocking; disturbance; passivity; self-sex; and stereotypy. Behavioral categories scored in experiments 3 and 4 are as follows: self-clasping; self-mouthing; huddling; locomotion; environment exploration; passivity; proximity; stereotypy; disturbance; and vocalizations.

The modified frequency scoring system consisted of 27 behavioral categories scored as to whether or not they occurred in each of 20 fifteen-second periods of which the five-minute testing session was comprised. The actual behavioral categories used in this system are shown below. This system, identical in the three studies (1, 2, and 3) in which it was used, is as follows: self-mouthing; self-biting; self-clasping; self-sex; passivity; proximity; huddling; rocking; grimace of fear; rejection; restraint; vocalness; stereotypy; locomotion; environment exploration; threat; approach; withdrawal; social contact; retrieving; grooming; attempted escape; ventral contact; and nonventral contact.

In all the behavioral scoring, objective written definitions of the behavioral categories were done prior to the start of actual data collection. Reliability scores between our trained observers on these scoring systems have yielded rank-order correlations of .95 or greater in all cases. This, in addition to the written definition of the behavioral categories, minimizes the possibility of both interobserver variations and variations due to subjective behavioral assessment.

In this report, clock-and-counter data will be presented from experiments 4 and 5, while modified frequency data will be presented from the first three studies. Direct comparison is, of course, not possible between modified frequency and clock-and-counter data, so all data were transformed into percentages of maximum possible scores so that some estimate of comparative levels might be achieved. Clock-and-counter raw data were all divided by 300, the number of seconds in an observation period, and modified frequency data were divided by 20, the maximum possible score for any behavior in that system. Statistical analyses were performed only within an experiment, using repeated measures analysis of variance on the raw scores. Transformation was done for graphic comparison only, and no statistical analyses were done across experiments or scoring systems.

In addition, data analyses for this particular report were restricted to data taken on the experimental (separation) groups, *after* separation only. In all cases, both control group and preseparation experimental group data were taken primarily for the purpose of verifying the essential "normality" of the mother-infant pairs in an undisturbed situation. Comparison of these data with those taken on separated infants is confounded by the difference in number of behaviors available to the infant in the two situations, ie, while with their mothers, the infants had maternal contact available, and indeed, spent a large amount of their test sessions interacting with their mothers (and peers in the case of experiment 4), while these behaviors were not available after separation. Thus, in this study, only postseparation experimental group data are presented for analysis.

Data from the five experiments are shown in Fig 1 to 3 and in the Table. For simplicity and clarity, we have presented only the data from the 4 PM, testing sessions for each study. Further, our



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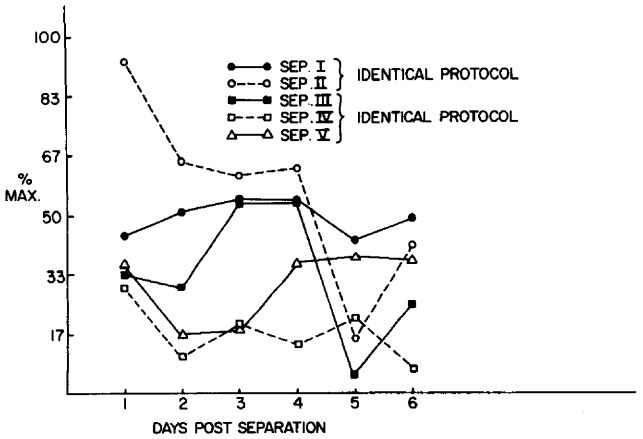



Fig 1.—Locomotion scores of rhesus infants following maternal separation (expressed as percent of maximum possible scores).

data consist of the first six postseparation days of each separation study, since this is the minimum length of separation in our studies and allows day for day comparisons.

The four behavioral categories considered here were chosen on the basis of two criteria: (1) relevance to detecting changes characteristic of the protest-despair model, and (2) consistency of definition between studies such that the comparison of data, while confounded by the difference between clock-and-counter and modified frequency scoring systems, is as meaningful as possible.

## RESULTS

### Locomotion

As an indicator of anaclitic depression, locomotion would be expected to be high in days 1 to 2 and much lower in the days following as the despair stage is reached. While data from experiment 2 would fit such an expectation, and in fact do show a significant ($P < .05$) decrease over days, the other studies are inconclusive. As an example, experiment 1, whose protocol was identical to that of experiment 2, and whose subjects were very close in age (5.8 months vs 6.3 months), showed an almost flat profile across the six days, at approximately 50% of the maximum possible score. Experiments 3, 4, and 5 also show widely varying data under identical protocols. While all three of these studies yielded similar scores for day 1 of approximately 33%, the patterns over days 2 to 6 are, qualitatively, quite different. Experiment 3 shows a rise and then a decrease; experiment 4 shows a drop to about 17% where it remains; and experiment 5 shows a decrease and then a rise. Changes over days were significant ($P < .05$) in experiments 1, 3, and 5, but not in 2 or 4. Thus, one of the measures often quoted as an indicator of the protest-despair changes shows considerable variability across experiments, both within identical protocols and across those two protocols we have used.

### Passivity

In a depressive response to separation, the predicted pattern for passive behavior would be an approximate inverse of locomotion—low during protest and higher in days 3 to 6 as despair appears. Again we show a great deal of variation across our studies. Comparison of experiments 1 and 2 (in which protocols and data acquisition were identical) shows quite different kinds of results. Experiment 2 again shows a curve along the general line of what we might expect in anaclitic depression, increasing across the six days, whereas experiment 1 data are not of the same shape, and change very little across time. The results under the identical protocols of experiments 3, 4, and 5 show a similar lack of congruity. Experiments 3 and 5 show rises from day 1 to 4, and then drop, but experiment 4 initially decreases, then rises, then falls again. Main effects for days are not significant ($P < .05$) except in experiment 2.

### Self-Clasping

This behavior would be expected to increase as despair occurs in days 3 to 6. Experiments 1 and 5 showed virtually no self-clasping throughout the study. Experiments 3 and 4 showed little or no self-clasping, except on day 3 when both rose to about 10% and then returned to very low levels. Experiment 2 showed no self-clasping until day 5 when it occurred about 45% of the time, after which levels returned to near zero. Day effects were significant ($P < .05$) for all studies at the .05 level by analysis of variance, but the heterogeneity of variance for zero scores vs all others confounds this statistic in this case.

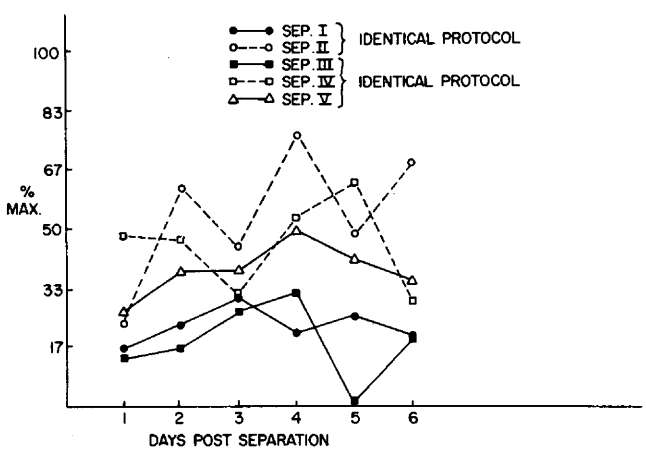



Fig 2.—Passivity scores of rhesus infants following maternal separation (expressed as percent of maximum possible scores).

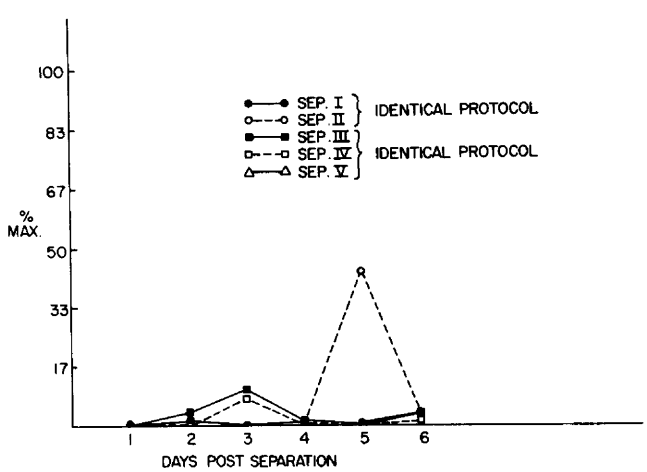



Fig 3.—Self-clasping scores of rhesus infants following maternal separation (expressed as percent of maximum possible scores).



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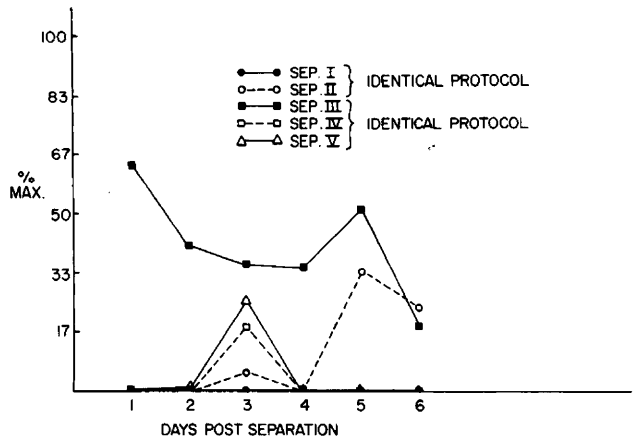


Fig 4.—Huddling scores of rhesus infants following maternal separation (expressed as percent of maximum possible scores).

## Huddling

This behavior would be expected to be a good indicator of despair. In experiment 3, this behavior occurred at very high levels for day 1 and gradually decreased through the rest of the six days, with a rise at day 5. Experiment 2 showed a rise on day 3 from zero, a subsequent increase from day 4 to 5, and a drop from day 5 to 6. Experiments 4 and 5, identical in protocol to experiment 3, showed a rise from low levels on day 2, to about 20% on day 3, and then dropped to zero. Experiment 1 showed no huddling throughout the study. Days effects were significant ($P < .05$) for the four studies in which huddling occurred, but again, the inherent heterogeneity of variance in comparing zero with nonzero scores makes interpretation difficult.

## COMMENT

Summary data from five studies of mother-infant separation in rhesus monkeys have been presented. While the comparisons made are on data taken by slightly different methods, measures of within-group changes are sufficiently valid to contrast the results of the five studies. These comparisons clearly do not support any unitary concept of a protest-despair response to mother-infant separation, at least not to the extent that such changes are predictable and stable enough to use this experimental paradigm as an induction method for producing animal models for neurobiologic or rehabilitative studies.

Several aspects of these studies should be considered for their possible contribution to the variability of response on the part of the infants. First, the ages of the infants varies from a mean of 5.9 months in experiment 1 to a mean of 8.5 months in experiment 4. Suomi et al[15] have noted differences between the responses to mother-infant separation of infants 60, 90, and 120 days of age, with animals separated at 90 days of age showing responses that might be most clearly interpreted in terms of the anaclitic depression model. In contrast, experiment 2 of the current studies showed the clearest protest-despair response, and the mean age of those subjects was 6.3 months. The younger subjects in experiment 1 (5.9 months) showed no biphasic activity change, and experiment 3, on infants 7.9 months of age, showed high levels of self-directed, withdrawn behaviors on day 1 of separation, although under a different protocol. Clearly, then, age alone does not define the severity nor kind of response to separation, at least in the studies discussed here.

The second factor that could be said to contribute to the wide variability of response to mother-infant separation in these studies may be the different protocols used. Experiments 1 and 2 were identical and involved moving the separated subjects to a new room where they could see and hear the other separation subjects, but no other animals. By contrast, experiments 3, 4, and 5 consisted of separating the infants from their mothers by a sheet of Plexiglas and preventing visual contact with any other animals. However, data from identical protocols (experiment 1 vs 2, and 3 vs 4 and 5) show wide divergence. Whereas, for example, locomotion remained virtually the same throughout experiment 1 and experiment 2 subjects showed high levels on day 1 and progressively lower levels through day 4. Likewise, note the virtually unchanging passivity levels in experiment 1 and the progressive rise in passivity levels for experiment 2. These two studies also used subjects of very close age and sex distribution. Comparison of experiments 3, 4, and 5 shows similar lack of consistency, again with animals of average ages within 1.4 months of each other.

It is virtually impossible to discuss the effects of sex of subjects within our studies. Subject availability in primate research is sometimes less than ideal, and these studies ended up being weighted toward male subjects (14 males and six females across all studies). Further research may indicate a contribution of subject sex to responses to mother-infant separation.

It becomes clear that mother-infant separation in rhesus monkeys does not fulfill the requirements of predictability and consistency, which are essential to establish a behaviorally induced model of human depression. It is of critical importance to have a stable behavioral paradigm to investigate criteria of human affective systems, and to investigate the biochemical basis for clinically effective treatments of those disorders, such as antidepressive drugs and electroconvulsive shock therapy. It is clear that consideration of other paradigms is essential to further progress in this area.

Bowden and McKinney[17] and Suomi and associates[18] have proposed peer separation as another model. For example, rhesus monkeys separated from their mothers at birth and reared together from age 15 days were subjected to 20 four-day periods of separation from each other from 90 to 300 days of age. The data from this study disclosed that the subjects showed clear protest and despair reactions to each separation, and that in fact, there was some evidence that the despair reactions became more intense as the number of separations increased. The authors suggested that this peer separation paradigm has advantages over the mother-infant separations. The same logic would apply, for our purposes, to the study of depressive disorders. First, the physical act of separating peers is less difficult than separating infants from mothers. Second, the nature of the mother-infant affectional system is normally a rapidly changing one during the first year of


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life, so that age of separation is of critical importance to that paradigm and large numbers of subjects of identical ages are necessary to generate meaningful data. Third, as an effect of the above, repeated observations in the same subjects of a phenomenon that lasts only a relatively short amount of time is not possible in a mother-infant separation paradigm. A fourth consideration of increasing importance in recent years is the relatively short period of time that mothers have to be involved in peer separation studies. Current primate resources are, for a variety of reasons, being limited more and more to within-colony production, and separation of the infants from their mothers at birth means that these mothers are that much more rapidly available for rebreeding purposes.

The paradigm of peer separation may have much to recommend it. If, as the data incidate, this technique does produce a more reliable, predictable, and stable separation response, in addition to being more efficient and powerful in terms of subject use, it would seem to be superior to mother-infant separation for purposes of animal research in human psychopathologic modeling. While other investigators have suggested its use[17-19] as a tool for studying affectional loss, perhaps resistance to using this technique as a model of human depressive disorders is caused by absence of analogous human deprivations, and the consequent lack of documentation of its relationship to the original models proposed by Spitz[1] and Robertson and Bowlby.[2]

Two of the major reasons for having animal models are (1) the investigation of the neurobiologic mechanisms of altered behavioral states, and (2) evaluation of the efficacy and mechanisms of therapeutic interventions. Repetitive peer separations offer the methodologic advantages of repeated treatments and the consequent potential for crossover designs. These advantages, plus the increased availability of subjects, when viewed in the context of the great inconstancy generated by mother-infant separation work, dictates that current and future research in separation models of depression should concentrate on the peer separation paradigm. This does not mean that mother-infant separation studies have no place. In helping to understand how the dynamics of mother-infant relationships and social organization influence infant development, including response to separation, these studies have been, and will continue to be, of great importance. This does not mean, however, that it is a useful paradigm for producing depression in animals.

Hopefully, more meaningful, consistent, and relevant information regarding depression can be gained using the peer separation paradigm. A closing word of caution, however: enthusiasm for a technique is often negatively correlated with the number of subjects studied, ie, low number of subjects leads to optimism. A relatively small number of subjects has been studied using the technique of peer separation, and although the results seem clear, the same was true with regard to mother-infant separation prior to the series of studies reported herein. Much more research on peer separation at many ages is needed to firmly establish both its usefulness for neurobiologic or rehabilitation studies and its validity as an animal model for human separation-induced depression.

Since this article was prepared, we have performed research that have yielded data potentially relevant to these studies. Isoniazid is routinely administered to all monkeys in our laboratory, and in many other primate laboratories, as a tuberculosis prophylactic drug. The known monoamine-oxidase inhibiting effects of this drug have been assumed by most investigators to be negligible. However, our recent research has indicated that animals subjected to peer separation while receiving isoniazid-supplemented chow behave differently than when the chow they are fed is not supplemented with isoniazid. These differences are suggestive of a more severe response to peer separations when animals are taken off isoniazid chow regimens, although replication in additional numbers of subjects is required. In addition, significant increases in activity of serum glutamate pyruvate transaminase levels have been found in animals as a function of removal of isoniazid from the chow. Although we think that the statements regarding the present research are not materially affected by these findings, the data showing effects of an heretofore ignored factor affecting response to separation and biochemical integrity of subjects in this laboratory should add another caveat to the interpretation of past and present primate depression literature.

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